



Calculating theoretical probabilities from two-way tables (two events)

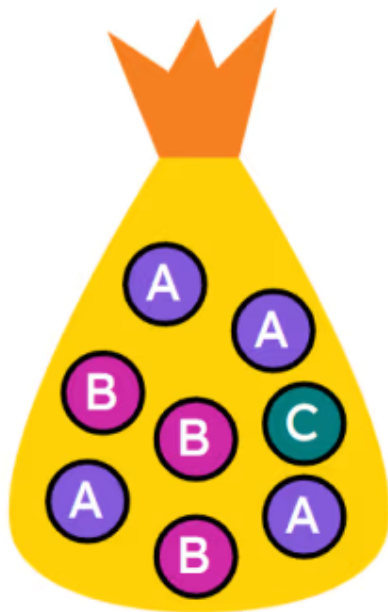
- 1 In a fair six-sided die, what is the probability of rolling a 5 and getting a head in a single toss of a fair coin? Tick **1** correct answer

☐ $\frac{1}{2}$

☐ $\frac{1}{6}$

☐ $\frac{1}{12}$

- 2 A counter is taken from the bag. Its letter is noted and then it is placed back into the bag. A counter is taken from the bag again. Which of these options is not the correct sample space? Tick **1** correct answer



☐ $\xi = \{AA, AB, AC, BA, BB, BC, CA, CB, CC\}$

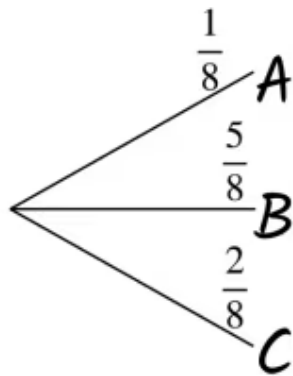
☐ $\xi = \{AA, AB, BA, BB, BC, CA, CB, CC\}$



- 3 If the probability of an event (A) is $\frac{3}{7}$, and the experiment is repeated 49 times, what is the theoretical number of times event (A) will occur? Tick **1** correct answer

- ☐ 3
☐ 7
☐ 21
☐ 49

- 4 Which of the events {A,B,C} is the least likely to occur according to this probability tree? Tick **1** correct answer



- ☐ A
☐ B
☐ C
☐ None.

- 5 John uses an unbiased random selection tool to generate a single letter of the alphabet. It is repeated three times. Which of the following letter sequences is the least likely? Tick **1** correct answer

- ☐ A,G,H
☐ A,A,A
☐ M,O,O
☐ B,I,N
☐ All selections are equally likely

- 6 There are two trials. In Trial 1, a spinner with {A, B} is spun twice. In Trial 2, a spinner with {B, C} is spun twice. Which statement is true? Tick **1** correct answer

- ☐ The total number of possible outcomes in Trial 1 is greater than in Trial 2.
☐ The total number of possible outcomes in Trial 1 is less than in Trial 2.
☐ The total number of possible outcomes in Trial 1 is the same as in Trial 2.



